Nonparametric intensity estimation in covariate-driven Poisson processes using deep learning

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Overview

★Covariate-driven Poisson processes

$Z \sim P_Z,\ X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$

- Covariates change the event intensity
- Estimating the intensity function $S(\cdot \mid \cdot)$ is crucial in applications
- We consider the theoretical property of the MLE of the intensity function

Example of covariate driven Poisson process



- Observations: $\{(Z_i, X_i)\}_{i=1}^n$

★Problem: The excess risk of MLE (KL div.) diverges easily
→ makes theoretical analysis difficult

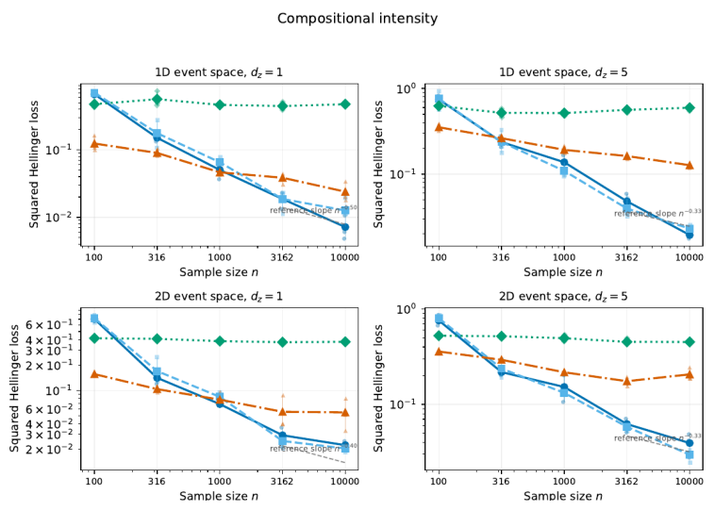
★Solution: Replace the KL divergence with the Hellinger distance

★ Derive an inequality that evaluates the performance of the MLD under a weaker assumption

★Show convergence rates for deep-learning-based MLE

★Prove the minimax optimality

★Numerical experiments



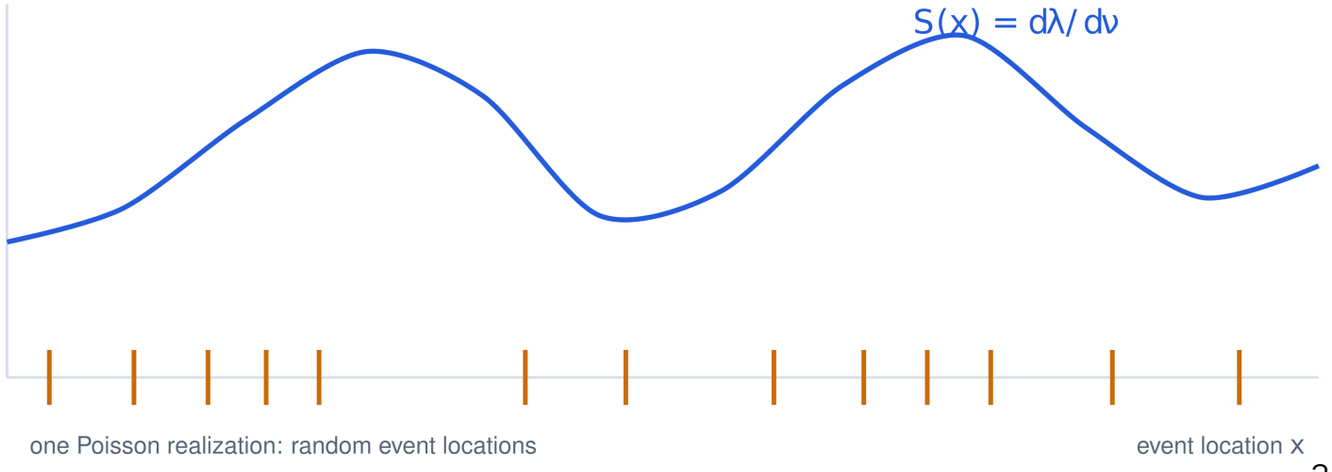
Poisson processes

Poisson Processes and Intensity Functions

- Poisson processes model random events occurring in time or space
- Key features of a Poisson process X
  - X is a non-negative integer-valued random measure; measures the number of events on a measurable set B
  - The numbers of events on disjoint regions are independent: $B_i \cap B_j = \emptyset \Rightarrow B_i \perp\!\!\!\perp B_j$
- The number of events on a measurable region follows the Poisson law: $X(B) \sim \mathrm{Poi}\{\int_B S(x)\, d\nu(x)\}$

Intensity function

- The intensity function controls expected counts and the distribution of events



Poisson Processes with Covariates

⭐Conditional intensity functions

- We consider intensity functions that vary depending on Z

$$X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$$

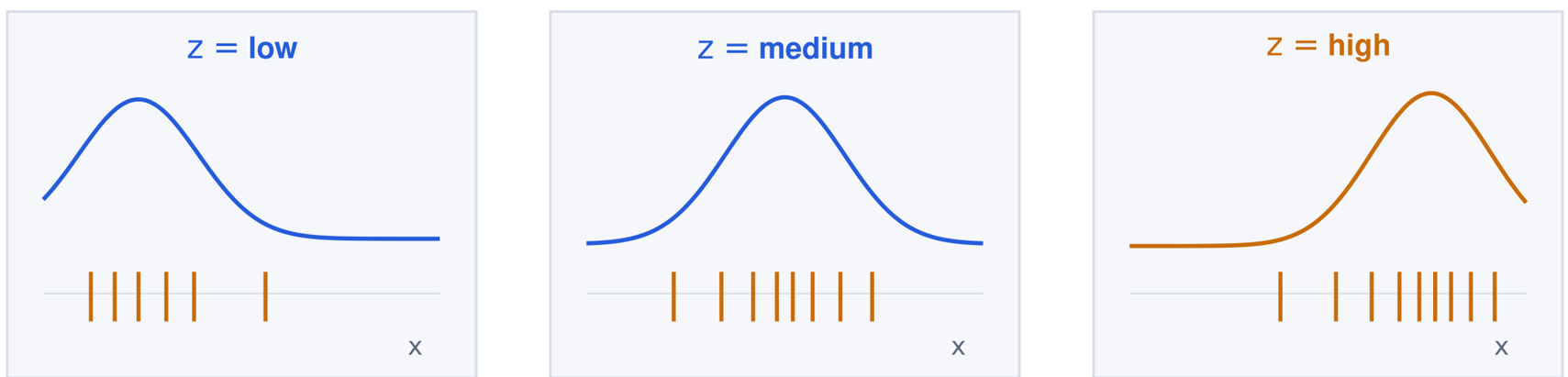
Conditional intensity function

⭐Why conditioning on Z?

- In real data, intensities naturally change due to many factors
- The intensity function may change its height, location, or shape with covariates Z

⭐Example: taxi pickups

- Location X = pickup site, Z = temperature, precipitation day type



Maximum Likelihood Estimation for Conditional Intensities

⭐Nonparametric maximum likelihood estimator (NPMLE)

- $A \subset \mathbb{R}^{d_x}$: space of events, $[0,1]^{d_z}$: space of covariates, $d = d_x + d_z$
- $(Z_1, X_1), \ldots, (Z_n, X_n)$: i.i.d. observations of covariates and events

$$Z_i \sim_{\text{i.i.d.}} P_Z, \quad X_i \mid Z_i \sim \text{Poisson process}(S_0(\cdot \mid Z_i))$$

- S_0: underlying (true) intensity function
- Negative log likelihood function:

$\boldsymbol{x}_{ij}$: events coresspondingto X_i
$N_i = X_i(A)$, ν: measure on A

$$\ell_n(X_1, \ldots, X_n, \boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \mid S) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \left\{ -\sum_{j=1}^{N_i} \log(S(\boldsymbol{x}_{ij} \mid \boldsymbol{z}_i)) + \int_A S(\boldsymbol{x} \mid \boldsymbol{z}_i)\, d\nu(\boldsymbol{x}) \right\}$$

- $\mathcal{F}_n$: hypothesis space
  - e.g., DNN, kernel-type estimator, etc…
- Nonparametric maximum likelihood estimator (NPMLE):

$$\hat{S}_n \in \min_{S \in \mathcal{F}_n} \ell_n(S)$$

- The integral can be approximated via Monte Carlo

Difficulties in Nonparametric Maximum Likelihood Estimation

⭐ Excess risk analysis of MLE in various settings

- **Regression**
  - If errors are Gaussian, the excess risk reduces to the squared error
  - If the true regression function is bounded, the squared error is finite
  - **This boundedness assumption is reasonable**
- **Classification**
  - The excess risk involves the **KL divergence** between the true and estimated conditional probabilities
    - **KL divergence is generally unbounded and can diverge easily**
  - **To evaluate the excess risk, we need additional assumptions**
    - e.g., the true conditional probability is bounded away from zero

$$c < \mathbb{P}(Y = 1 \mid \boldsymbol{x}) < C, \quad 0 < c, C < 1$$

- **Estimating intensity functions**
  - **Just like classification, KL divergence diverges easily**

Difficulties in Nonparametric Maximum Likelihood Estimation

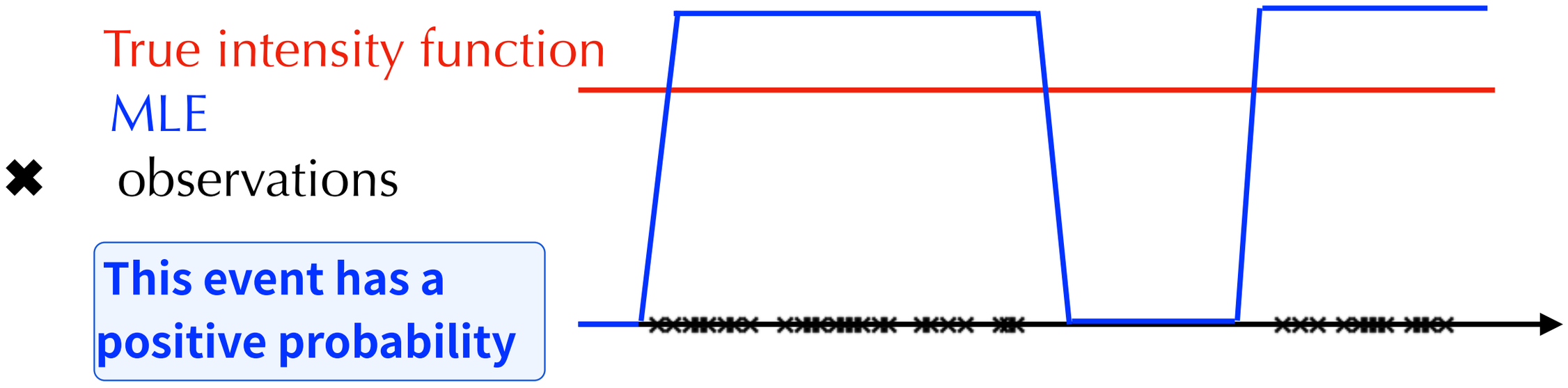
⭐Example where the KL divergence diverges

- For Poisson processes, the KL divergence involves **the ratio of intensity functions**:

$$\mathrm{KL}(P_f, P_g) = \int_A \left(f \log\left(\frac{f}{g}\right) + g - f \right) d\nu$$

P_h : the distribution of a Poisson process with intensity function h

- If the support of the estimator does not cover the support of the true intensity function, the KL divergence becomes infinite



Related Works

⭐Related works

1 Intensity estimation

Kernel, thresholding, Bayesian GP/RKHS, and manifold estimators.
Diggle (1985); Kirichenko & van Zanten (2015) Reynaud-Bouret & Rivoirard (2010); Ward et al. (2023)

2 Covariate-driven modeling

Parametric, kernel, and applied deep models with covariates.
Lawless (1987); Klutchnikoff & Massiot (2016); Okawa et al. (2019); Wang et al. (2020)

3 Closest recent theory

Bayesian growing-domain theory;

Giordano et al. (2026)

4 General theory

Hellinger M-estimation; DNN rates for compositional and manifold structure.
van de Geer (2000); Schmidt-Hieber (2020); Bos & Schmidt-Hieber (2022); Chen et al. (2022)

→

Gap addressed here

Joint theory for

- replicated conditional Poisson data
- admit the true intensity vanishes
- near-minimax rates
- Deep-learning-based NPMLE

To our knowledge, these ingredients have not been analyzed jointly for conditional Poisson NPMLE.

Goals of This Study

⭐To clarify the theoretical properties of the NPMLE of intensity functions for Poisson processes with covariates

- Issue: KL divergence diverges easily → standard analysis fails
- Approach: Evaluate Hellinger distance instead

⭐Analyze deep-learning–based NPMLE

- If the true intensity is composite structured
  → avoids curse of dimensionality, achieves nearly optimal rate
- If supported on a Riemannian manifold
  → automatically adapt to low-dimensional
  Riemannian manifold structures

Notations & Assumptions

⭐Notations

- $\tilde{H}^2(S_1, S_2) := \mathbb{E}_Z \left[\frac{1}{2} \int_A (\sqrt{S_1(\boldsymbol{x} \mid Z)} - \sqrt{S_2(\boldsymbol{x} \mid Z)})^2 d\boldsymbol{x} \right]$
  - **Instead of KL divergence between Poisson processes, we evaluate $\tilde{H}^2$**
- $\mathcal{F}(L, r, B, F)$: class of DNN
  - L : depth, r : width per layer, B : parameter magnitude bound
  - F : bound of the output
  - Activations: ReLU everywhere, including the output layer

⭐Assumption

$$\exists c_0 > 0,\ \forall n \in \mathbb{N},\ \exists S_n^* \in \mathcal{F}_n,\ \forall \boldsymbol{x} \in A,\ \forall \boldsymbol{z} \in C;\ \frac{S_0(\boldsymbol{x} \mid \boldsymbol{z})}{S_n^*(\boldsymbol{x} \mid \boldsymbol{z})} \leq c_0^2$$

- This holds automatically when using the DNN estimator

Solution: Direct Hellinger Control

⭐Key idea: technique from van de Geer (2000)

- R(S): KL risk

$$R(S) \geq 2\widetilde{H}^2(S, S_0),$$
$$R(\widehat{S}_n) = \infty.$$

∞ above Hellinger gives no upper control.

1 **Likelihood optimality** $\ell_n(\widehat{S}_n) \leq \ell_n(S_n^\star).$

2 **Midpoint stabilizes the log likelihood ratio** $M_n := \frac{\widehat{S}_n + S_n^\star}{2}, \qquad \log \frac{M_n}{S_n^\star} \geq -\log 2.$

3 **Basic inequality with respect to the Hellinger distance**

$$\tilde{H}^2(M_n, S_n^\star) \lesssim \text{empirical process term} + \text{approximation error}$$

⭐Poisson-specific difficulty: the lower bound is random

Total lower bound $-\frac{N_i}{2}\log 2$

N_i is a Poisson random variable and unbounded.

Condition on N_i

↓

Poisson MGF ⇒ Bernstein-size control

Oracle Inequality

★Oracle inequality

Theorem1 (Oracle Inequality)

We consider the estimation of the conditional intensity function of a Poisson process. Suppose that δ_n and a universal constant c satisfies the following inequality:

$$\sqrt{n}\delta_n^2 \geq c\Psi(\delta_n)$$

Then, for any $\delta \geq \delta_n$, the following inequality holds.

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \leq 1224(c_0+1)^2 \left(\underbrace{\delta^2}_{\text{Variance}} + \frac{c^3}{n} + \underbrace{\tilde{H}^2(S_n^*, S_0)}_{\text{Bias}^2} \right)$$

- Ψ: model complexity
- The NPMLE convergence rate depends on
  the Complexity (Variance) term + the Approximation (Bias) term

Oracle Inequality

⭐The upper bound depends on complexity and approximation error

- The upper bound, ignoring constants:

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \lesssim \delta^2 + \frac{c^3}{n} + \tilde{H}^2(S_n^*, S_0)$$

1 Complexity

$$\sqrt{n}\,\delta_n^2 \gtrsim \Psi(\delta_n).$$

Ψ measures the local bracketing entropy of the model class.

2 Approximation

$$S_n^\star \in \mathcal{F}_n, \qquad \frac{S_0}{S_n^\star} \le c_0^2.$$

S_n^* may depend on the unknown truth; choose it as close to S_0 as possible.

Complexity ↑ → **Richer model class** → **Approximation ↓**

⭐Choose architecture so that the two leading terms have the same order (when model is DNN)

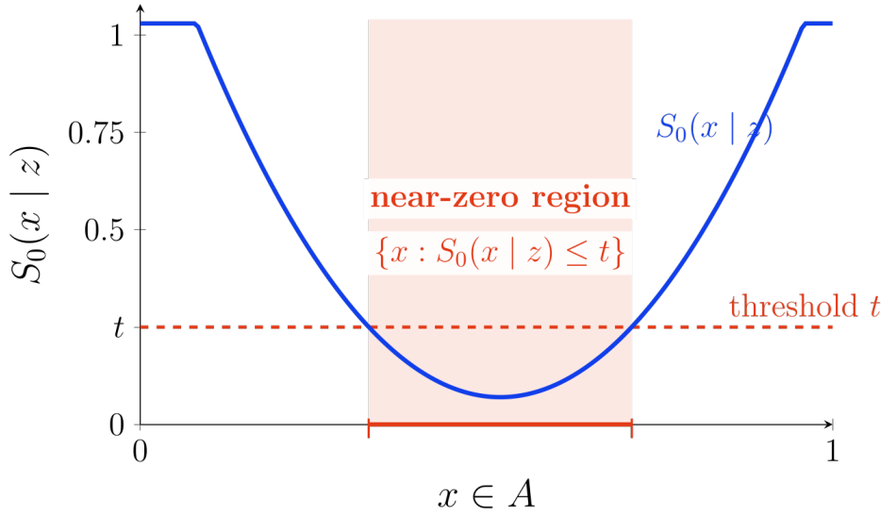
Small Value Bound (SVB)

⭐Small Value Bound condition improves approximation error

Definition

$$V_S(t) := \int_A \mathbb{P}_Z(S(x \mid Z) \le t)\, d\nu(x) \le Ct^{\alpha}, \qquad t > 0.$$

Larger α ⇒ the near-zero region shrinks faster



Fixed z; SVB averages its volume over Z

What SVB gives at the clipping scale t = M⁻¹

For an M⁻¹-accurate approximation clipped below at M⁻¹:

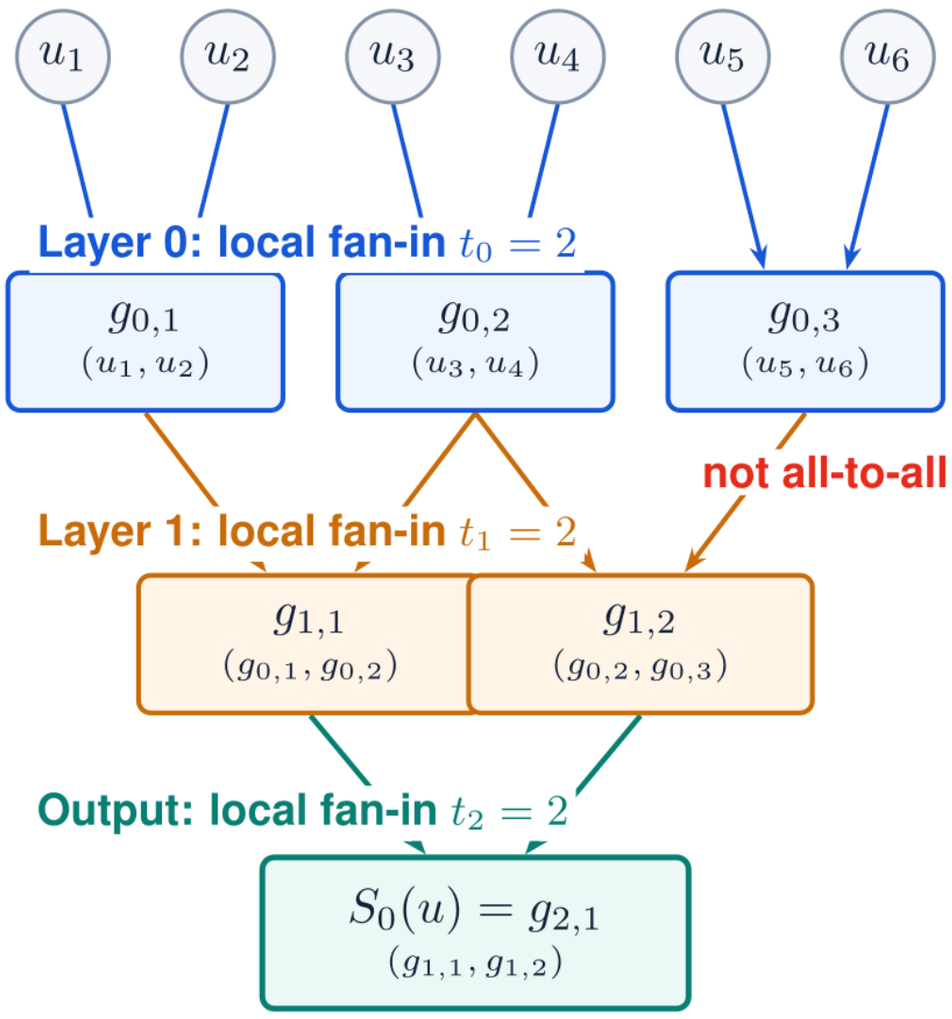
$$V_{S_0}(M^{-1}) \le CM^{-\alpha} \quad \Longrightarrow \quad \widetilde{H}^2(S, S_0) \lesssim M^{-\{1+(\alpha\wedge 1)\}} \log M.$$

α = 0: no restriction | α > 0: no positive-volume zero set | rate gain saturates at α = 1

Structure of the True Intensity Function

⭐Composition structure

Sparse dependency graph



Effective dimension t_i = number of active predecessors
Example: $t_0 = t_1 = t_2 = 2 \ll d_0 = 6$.

Definition of $\mathcal{G}_{\text{comp}}$

$\mathcal{C}^\beta(D, Q)$: β-Hölder functions on D with Hölder norm $\leq Q$.

$$\mathcal{G}_{\text{comp}}(q, \mathbf{d}, \mathbf{t}, \boldsymbol{\beta}, Q) := \big\{ f = g_q \circ \cdots \circ g_0 \geq 0 : g_i = (g_{ij})_{j=1}^{d_{i+1}} : [a_i, b_i]^{d_i} \to [a_{i+1}, b_{i+1}]^{d_{i+1}}, \; g_{ij} \in \mathcal{C}^{\beta_i}([a_i, b_i]^{t_i}, Q), \; i = 0, \ldots, q, \quad |a_i| \vee |b_i| \leq Q \big\}.$$

$\mathbf{d} = (d_0, \ldots, d_{q+1})$, $d_{q+1} = 1$; $\mathbf{t} = (t_0, \ldots, t_q)$; $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_q)$.
t_i: maximum number of active variables in a component g_{ij}.

Effective smoothness

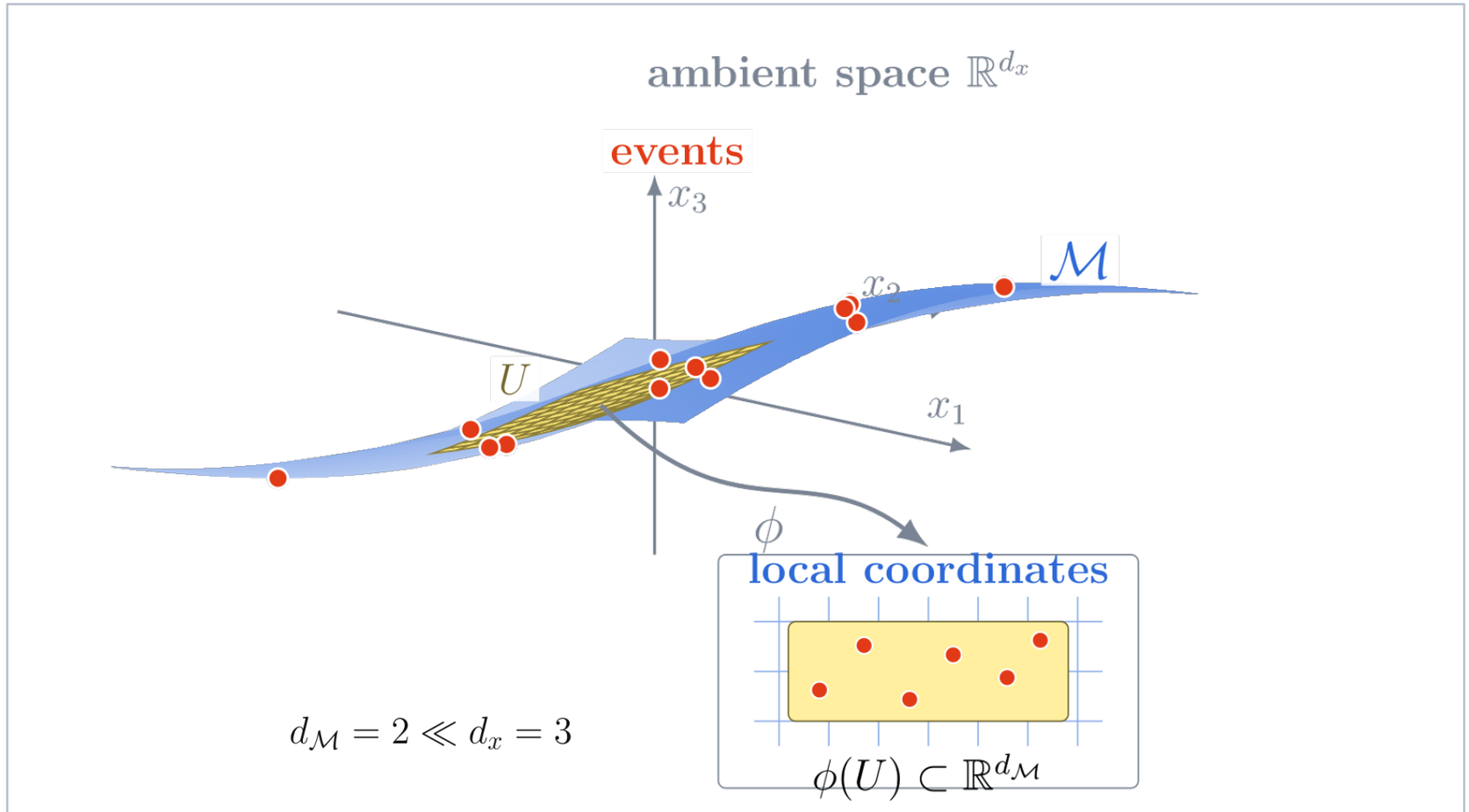
$$\beta_i^* := \beta_i \prod_{\ell=i+1}^{q} (\beta_\ell \wedge 1), \qquad i = 0, \ldots, q.$$

The empty product is 1, so $\beta_q^* = \beta_q$. Outer maps with $\beta_\ell < 1$ attenuate inner smoothness; difficulty is indexed layerwise by (t_i, β_i^*).

Structure of the True Intensity Function

⭐Manifold supported

- Intensity supported on a low-dimensional manifold
  - $\mathcal{C}^{\beta}(\mathcal{M} \times [0,1]^{d_z})$
  - $\mathcal{M} \subset A$ is a compact Riemannian manifold with $d_{\mathcal{M}} < d_x$
    - $\dim(\mathcal{M}) = d_{\mathcal{M}}$



Convergence Rates for Deep Learning Estimators

★Convergence rates for DNN-based MLE

- Using Theorem 1, we obtain convergence rates for fixed function classes of the true intensity function

When $\alpha = 0$

function class	$\mathcal{C}^{\beta}(\mathcal{M} \times [0,1]^{d_z}, Q)$	$\mathcal{G}_{\text{comp}}$
convergence rates (exponent of n)	$-\frac{\beta}{\beta + d_{\mathcal{M}} + d_z}$	$-\max_{i=0,\ldots,q} \frac{\beta_i^*}{\beta_i^* + t_i}$

- **Notably, the convergence rates do not depend on the ambient dimension d**

★Why is the exponent different from the classical rate?

If S_0 is bounded away from zero

$$S_0 \geq c > 0: \quad (\sqrt{S} - \sqrt{S_0})^2 \lesssim |S - S_0|^2$$

If S_0 may equal zero

$$S_0 = 0: \quad (\sqrt{S} - \sqrt{S_0})^2 = |S - S_0|$$

At zeros, Hellinger approximation error is not quadratic.

$$\alpha = 0: \ n^{-\frac{\beta}{\beta + d_{\text{eff}}}} \quad \Longleftrightarrow \quad \alpha \geq 1: \ n^{-\frac{2\beta}{2\beta + d_{\text{eff}}}}$$

Minimax Optimality

⭐Minimax lower bound

Benchmark rate

$$r_n := \max_{i=0,\dots,q} n^{-\beta_i^*/(\beta_i^*+t_i)},$$
$$\beta_i^* := \beta_i(\beta_{i+1} \wedge 1) \cdots (\beta_q \wedge 1).$$

Each t_i is the active input dimension of g_i

Minimax risk

$$R_n(\mathcal{G}) := \inf_{\widetilde{S}_n} \sup_{S_0 \in \mathcal{G}} \mathbb{E}_{S_0}\left[\widetilde{H}^2(\widetilde{S}_n, S_0)\right].$$

Smallest worst-case risk over all estimators

Bounds match for the composition class (α = 0)

$$c\, r_n \;\le\; R_n(\mathcal{G}_{\text{comp}}) \;\le \sup_{S_0 \in \mathcal{G}_{\text{comp}}} \mathbb{E}_{S_0}\left[\widetilde{H}^2(\widehat{S}_n, S_0)\right] \;\le\; C\, r_n \log^3 n.$$

No estimator is uniformly faster

DNN-NPMLE attains the same exponent

- The convergence rate derived in the previous slide matches this (up to log-factor) → **Our rate is minimax optimal**

Numerical Experimtents

⭐Design

$$Z_i \sim \mathrm{Unif}([0,1]^{d_z}), \qquad X_i \mid Z_i = z_i \sim \mathrm{PP}(S_0(\cdot \mid z_i)), \quad i = 1, \ldots, n.$$

PP = Poisson point process

Ground truth

- compositional intensity
- manifold support: S^1 or S^2

Estimators

- DNN–Softplus
- DNN–ReLU
- kernel estimator

Setting

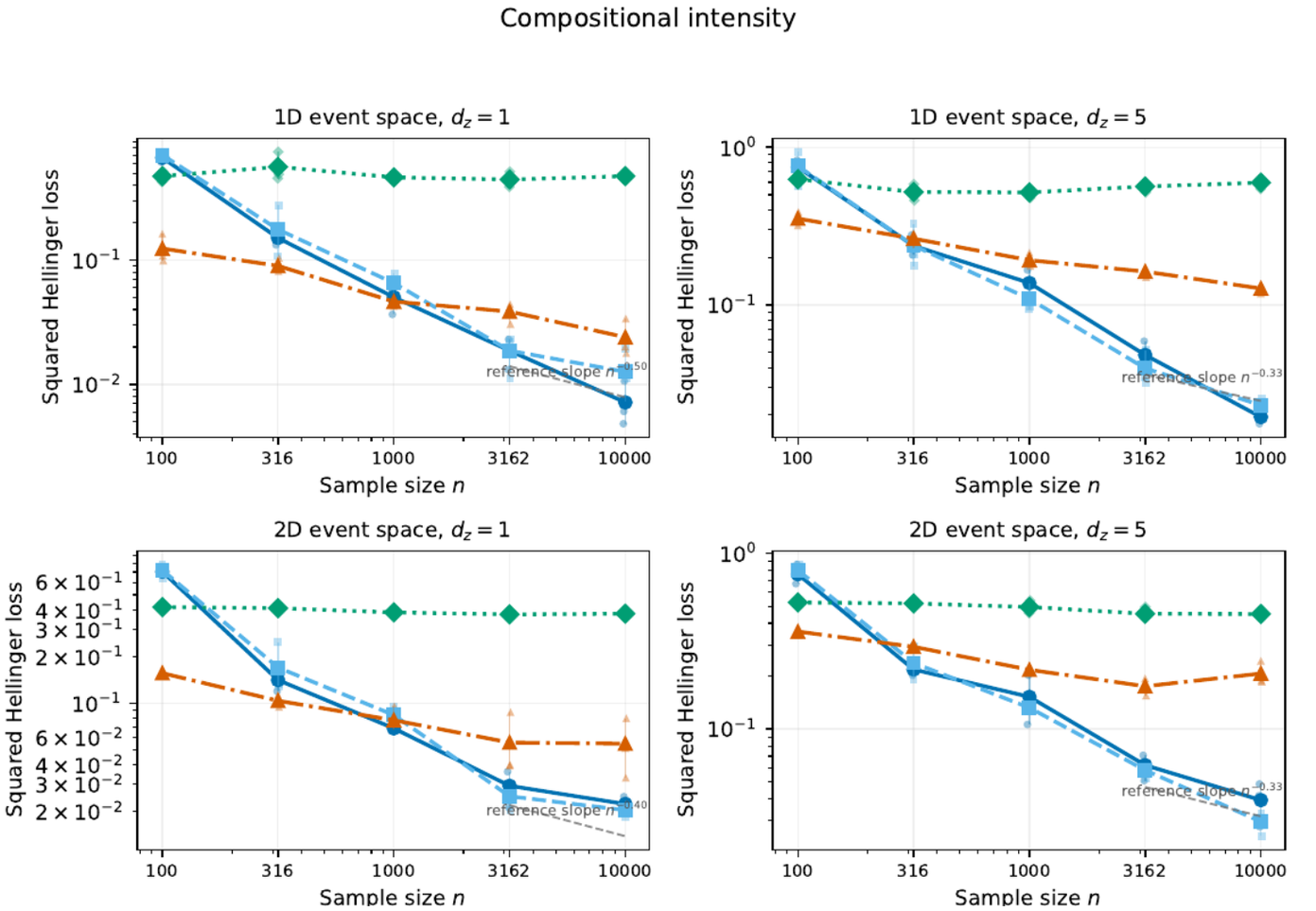
n ∈ {100, 316, 1000, 3162, 10000}
d_z ∈ {0, 1, 5}
E[events per replicate] ≈ 30

Evaluation: covariate-averaged squared Hellinger loss

$$\widetilde{H}^2(\widehat{S}, S_0) := \mathbb{E}_Z \left[\int_A \left(\sqrt{\widehat{S}(x \mid Z)} - \sqrt{S_0(x \mid Z)} \right)^2 d\nu(x) \right].$$

Numerical Experimtents

⭐Compositional functions



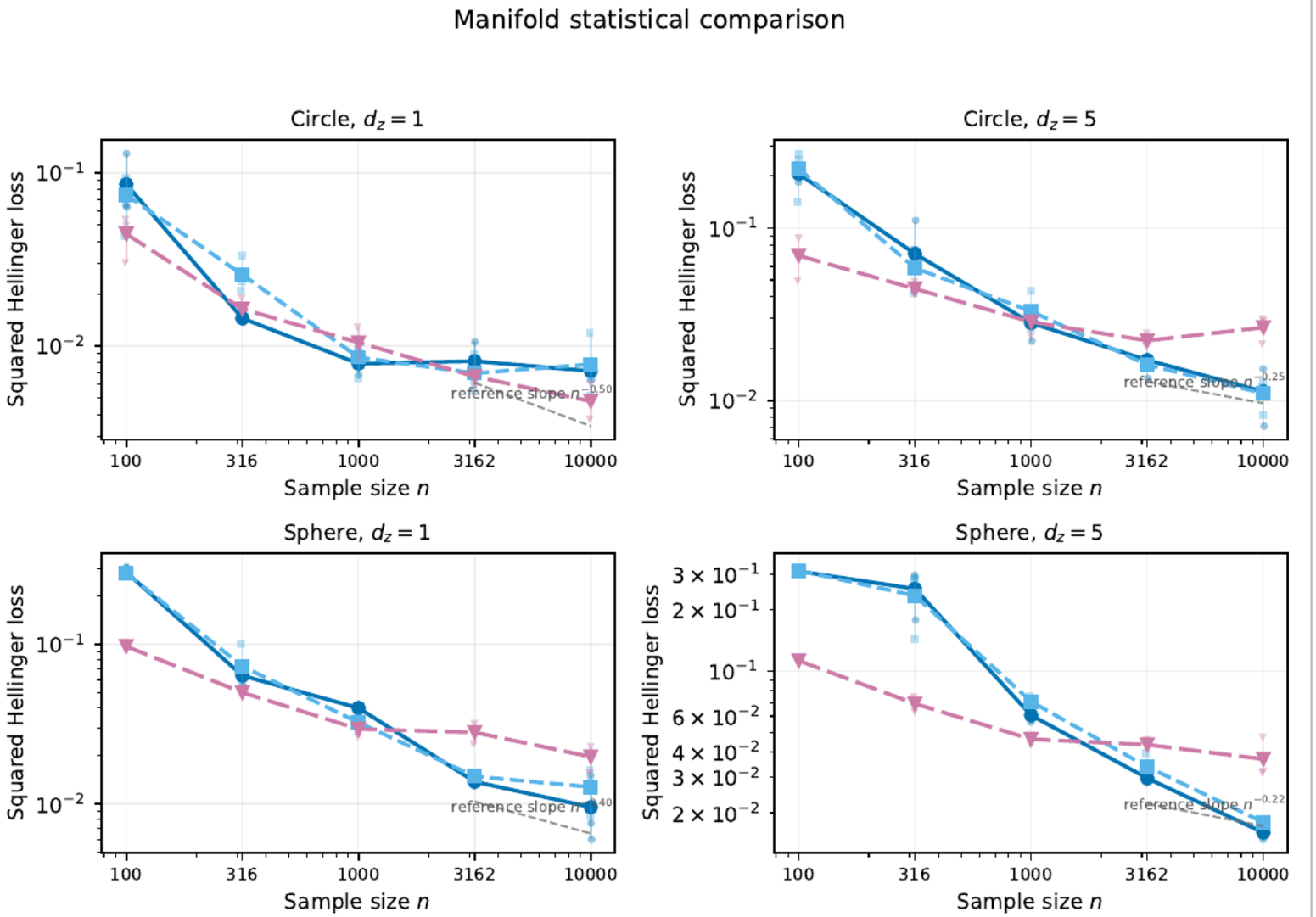
Mean loss at n = 10,000

Setting	Best DNN	Conditional kernel
1D d_z = 1	**0.00716**	0.0239
1D d_z = 5	**0.0195**	0.127
2D d_z = 1	**0.0205**	0.0548
2D d_z = 5	**0.0297**	0.207

DNN is best in all four panels

Numerical Experimtents

⭐Manifold supported



Mean loss at n = 10,000

Setting	Best DNN	Oracle kernel
Circle d_z = 1	0.00715	**0.00478**
Circle d_z = 5	**0.0110**	0.0265
Sphere d_z = 1	**0.00953**	0.0197
Sphere d_z = 5	**0.0160**	0.0368

Oracle kernel wins only the easiest case

circle with d_z = 1

Summary

- ★**We establish theoretical guarantees for NPMLE of covariate-driven Poisson intensities (Theorem 1)**

- ★**We derive convergence rates for deep learning-based MLE**
  - Deep learning can avoid the curse of dimensionality
  - Minimax optimality is proved

function class	$\mathcal{C}^\beta(\mathcal{M} \times [0,1]^{d_z}, Q)$	$\mathcal{G}_{\mathrm{comp}}$
convergence rates (exponent of n)	$-\dfrac{\beta}{\beta + d_{\mathcal{M}} + d_z}$	$-\max\limits_{i=0,\ldots,q} \dfrac{\beta_i^*}{\beta_i^* + t_i}$

Minimax optimal

Appendix

Sparsity in DNN

⭐Sparsity in DNN

- Many papers assume neural network sparsity,
  i.e., few nonzero parameters
  - 📌 e.g., Suzuki (2019), Schmidt-Hieber (2020)
- Why? ⇒ excess risks are typically bounded by:

$$\frac{\text{“Complexity”}}{n} + \text{Bias}^2$$

- “Complexity”: covering number, bracketing number
- Covering number scales with the number of parameters
  → Sparsity leads to a tighter bound
- **Using tighter bias control (Kohler & Langer, 2021), we achieve fast rates without requiring sparsity**